

Feature-based Interpretation of spectral clustering solutions for mixed data

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Outline

Spectral clustering in a nutshell

From pairwise distances to a graph partition

Feature-aware distances for mixed data

Which numerical and categorical differences enter the graph?

Association- and interaction-aware construction

Encoding redundancy and continuous–categorical structure

Synthetic evidence and interpretation

What spectral clustering can recover from each representation

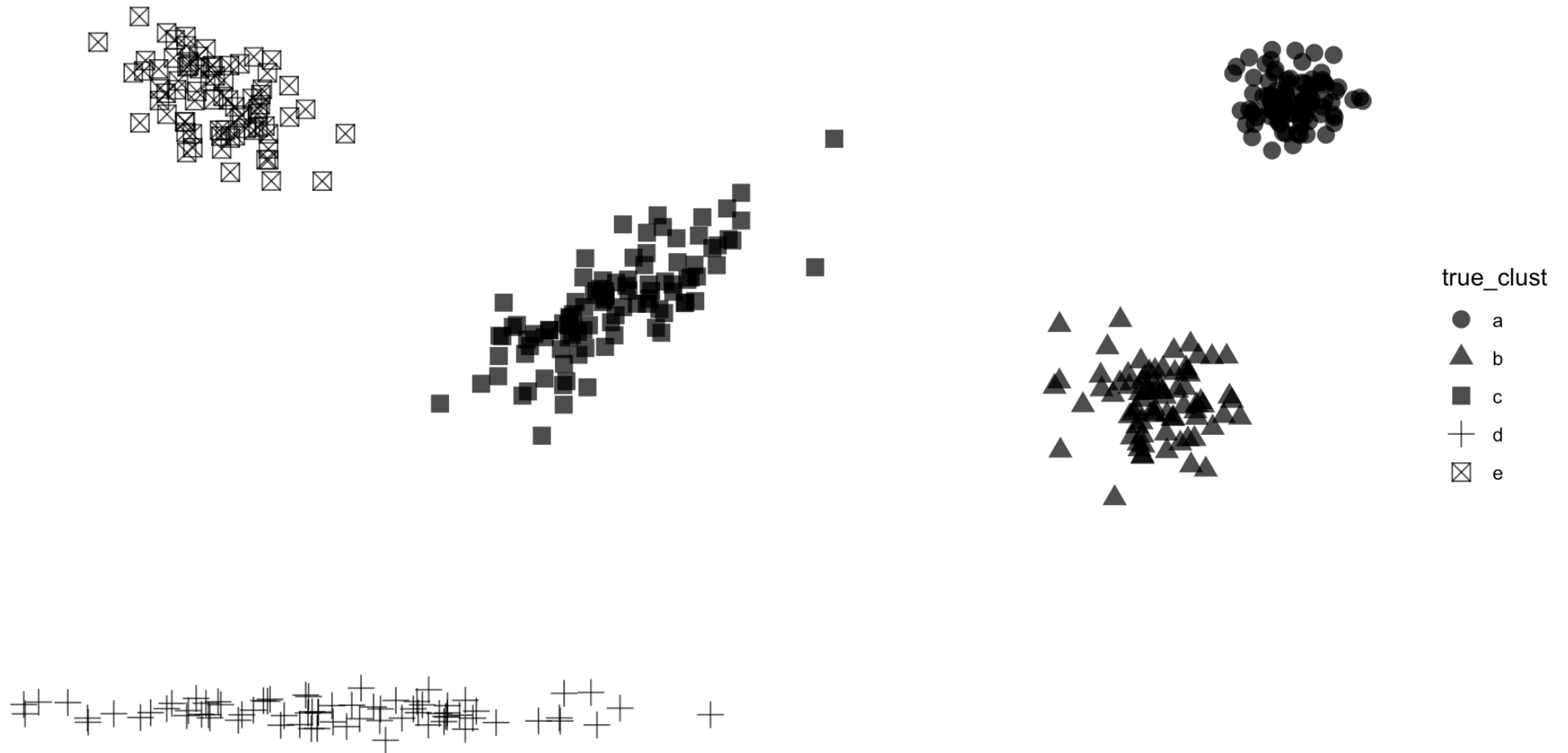
cluster analysis

the general aim is to find homogeneous groups of observations

- observations from the same group are similar to each other
- observations from different groups are not similar to each other
- multiple clustering approaches exist

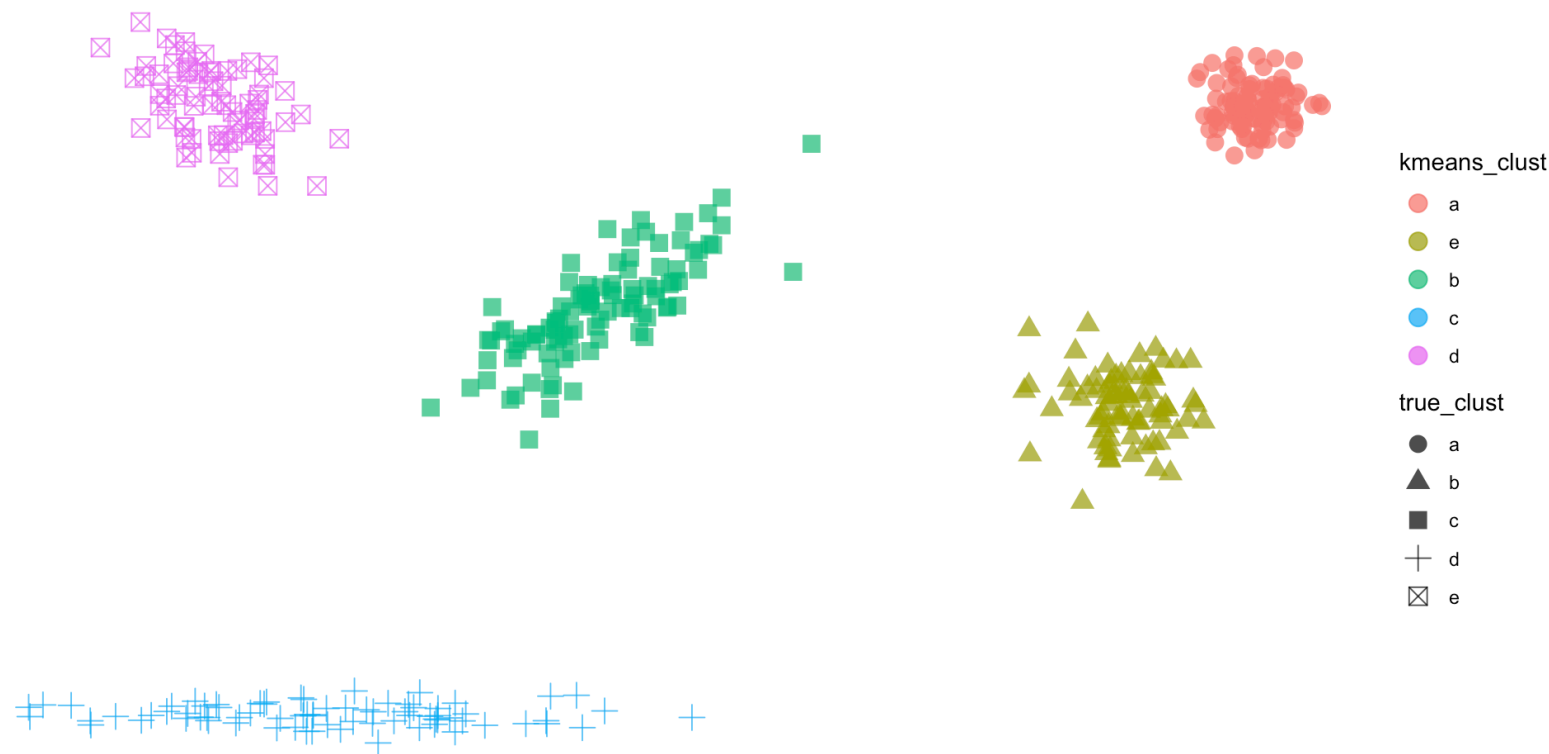
clustering approaches

- If a cluster structure does underlie the observations at hand, any approach will work



clustering approaches

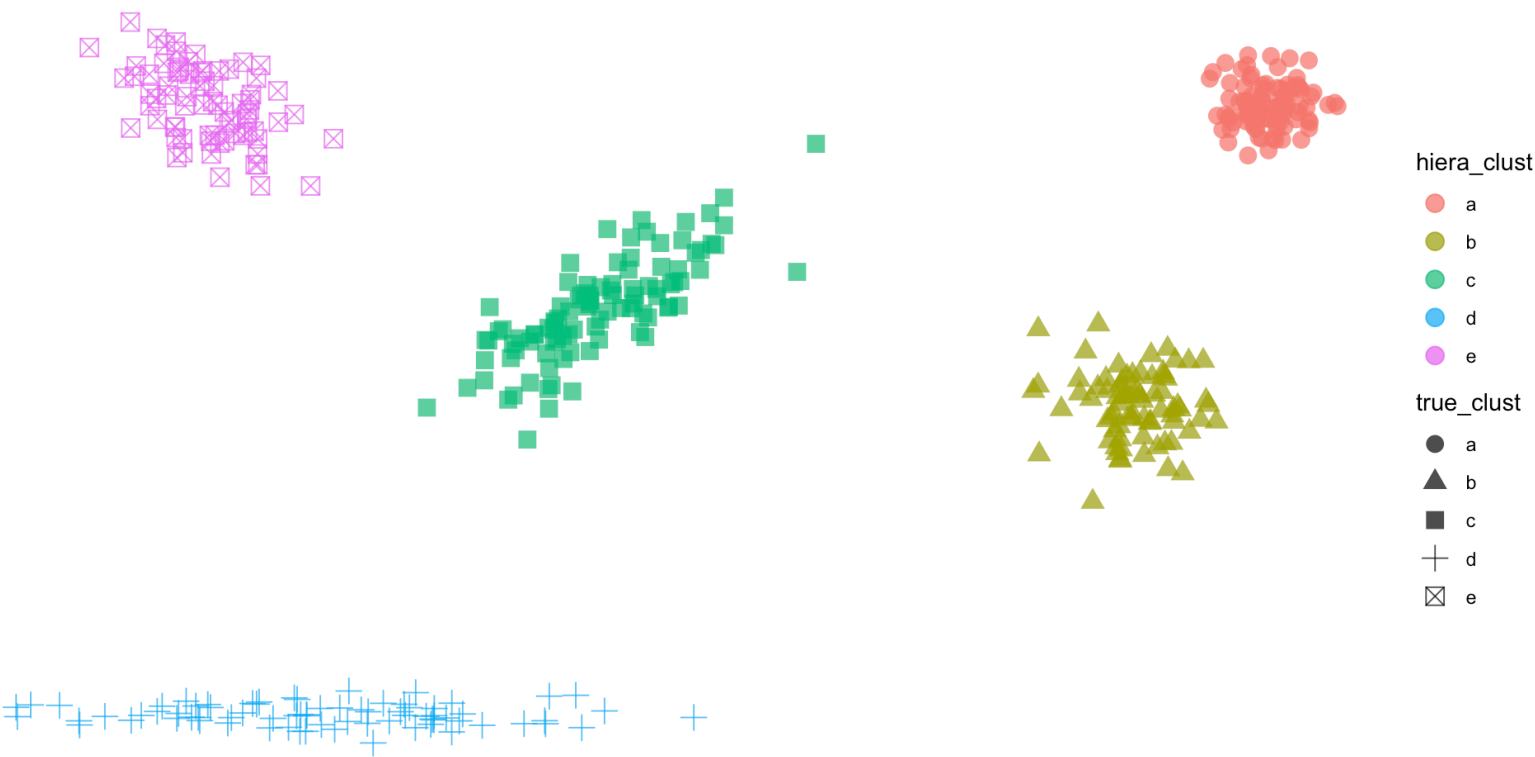
partitioning¹



1. MacQueen’s K -means; MacQueen (1967)

clustering approaches

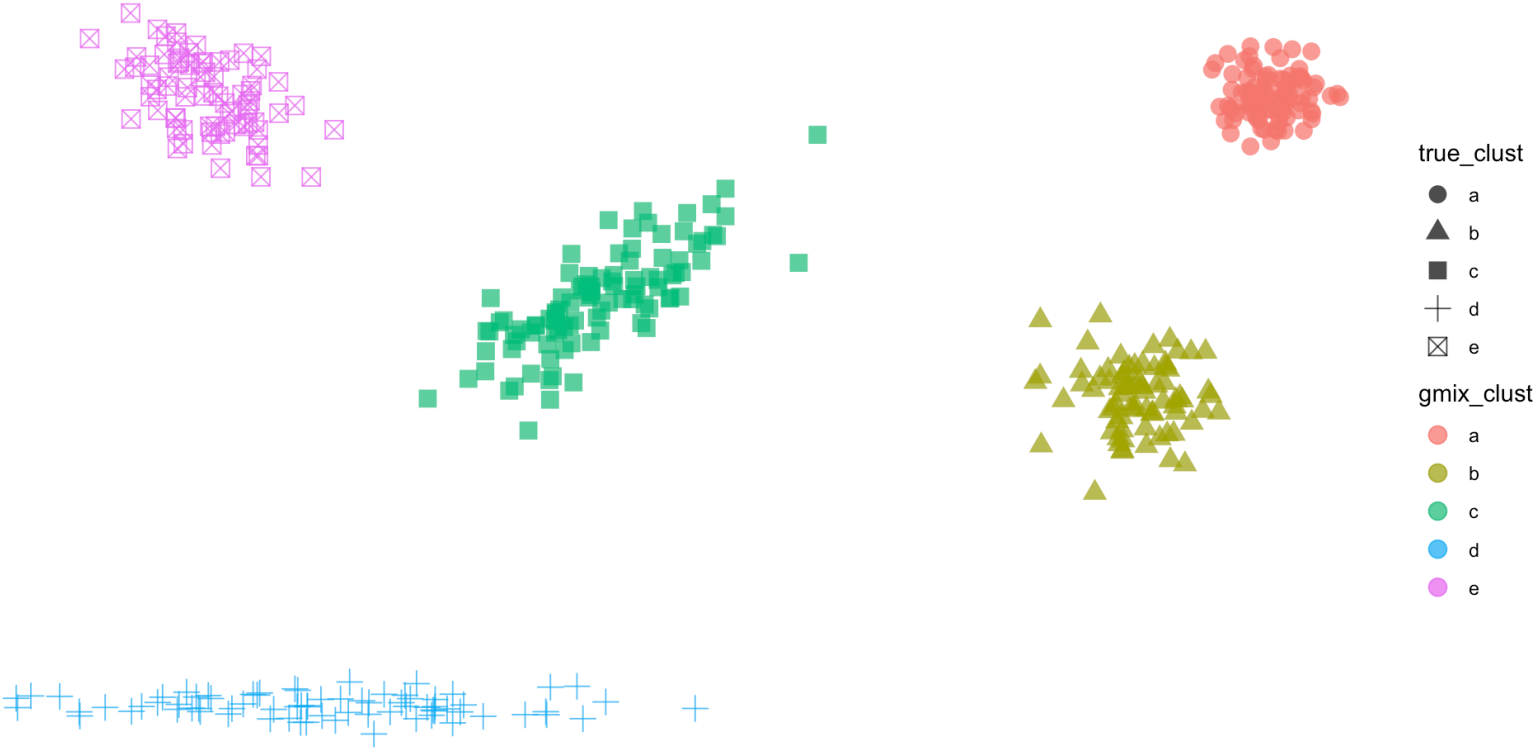
agglomerative¹



1. Kaufman & Rousseeuw (2009)

clustering approaches

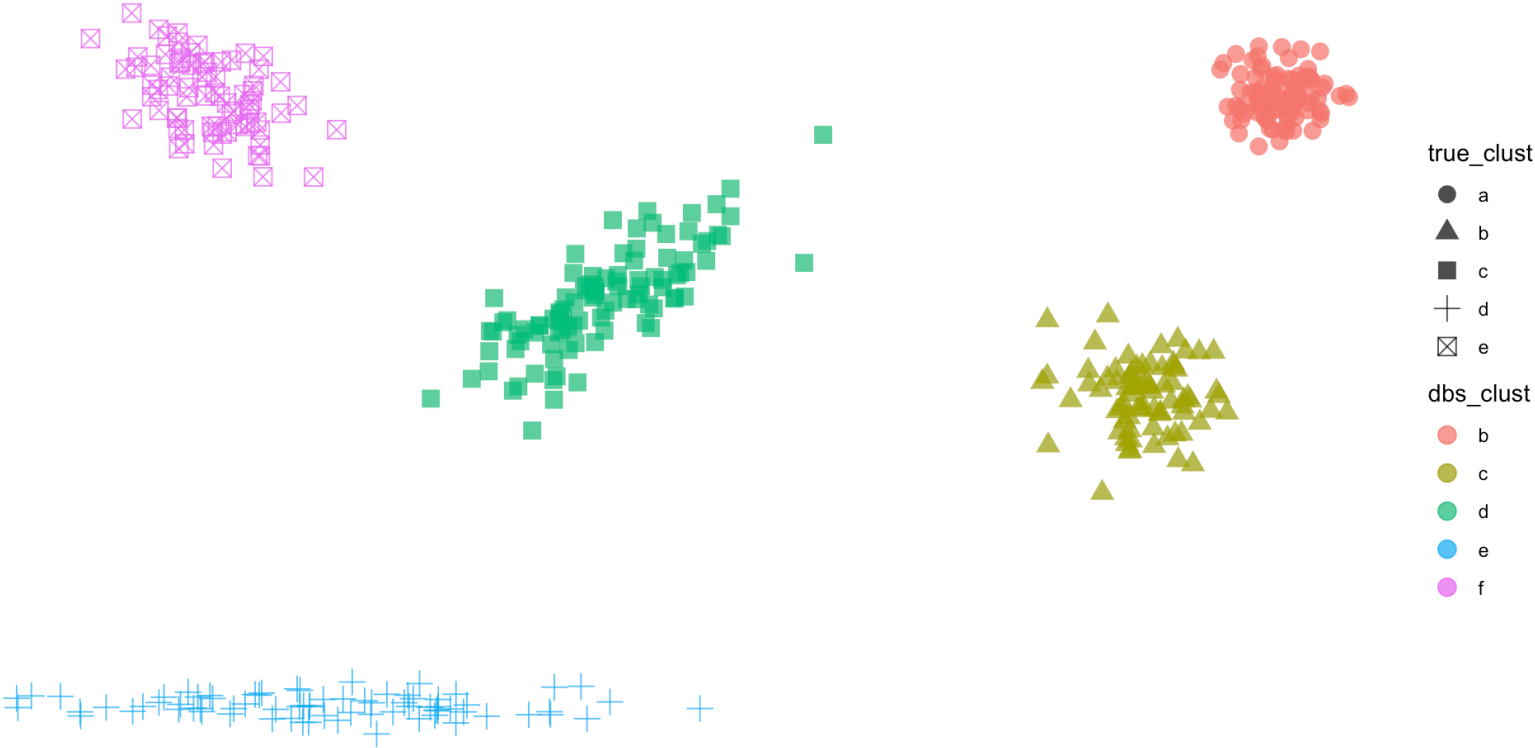
model-based¹



1. McLachlan & Basford (1988)

clustering approaches

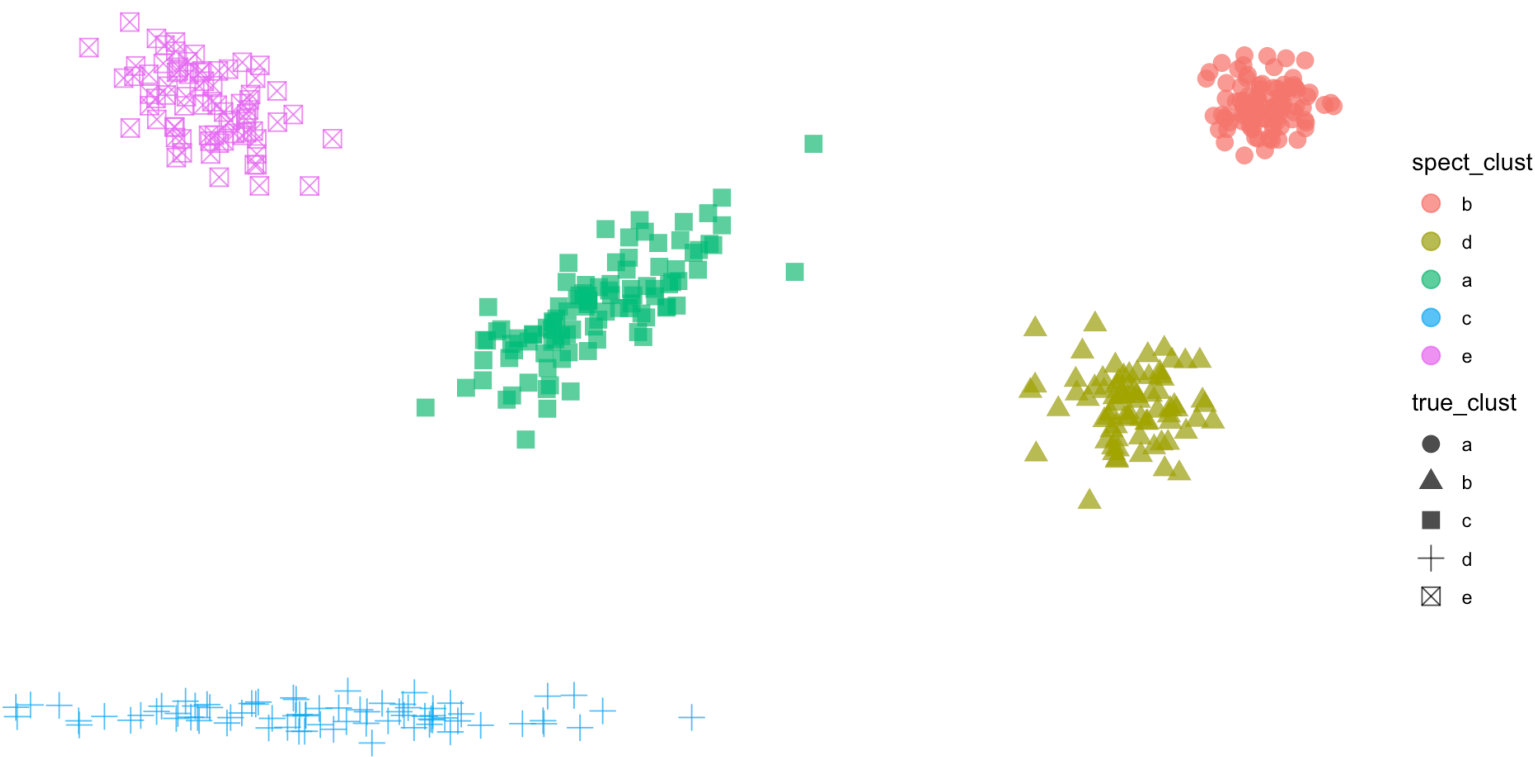
density-based¹



1. DBSCAN; Ester, Kriegel, Sander, Xu, et al. (1996)

clustering approaches

spectral¹

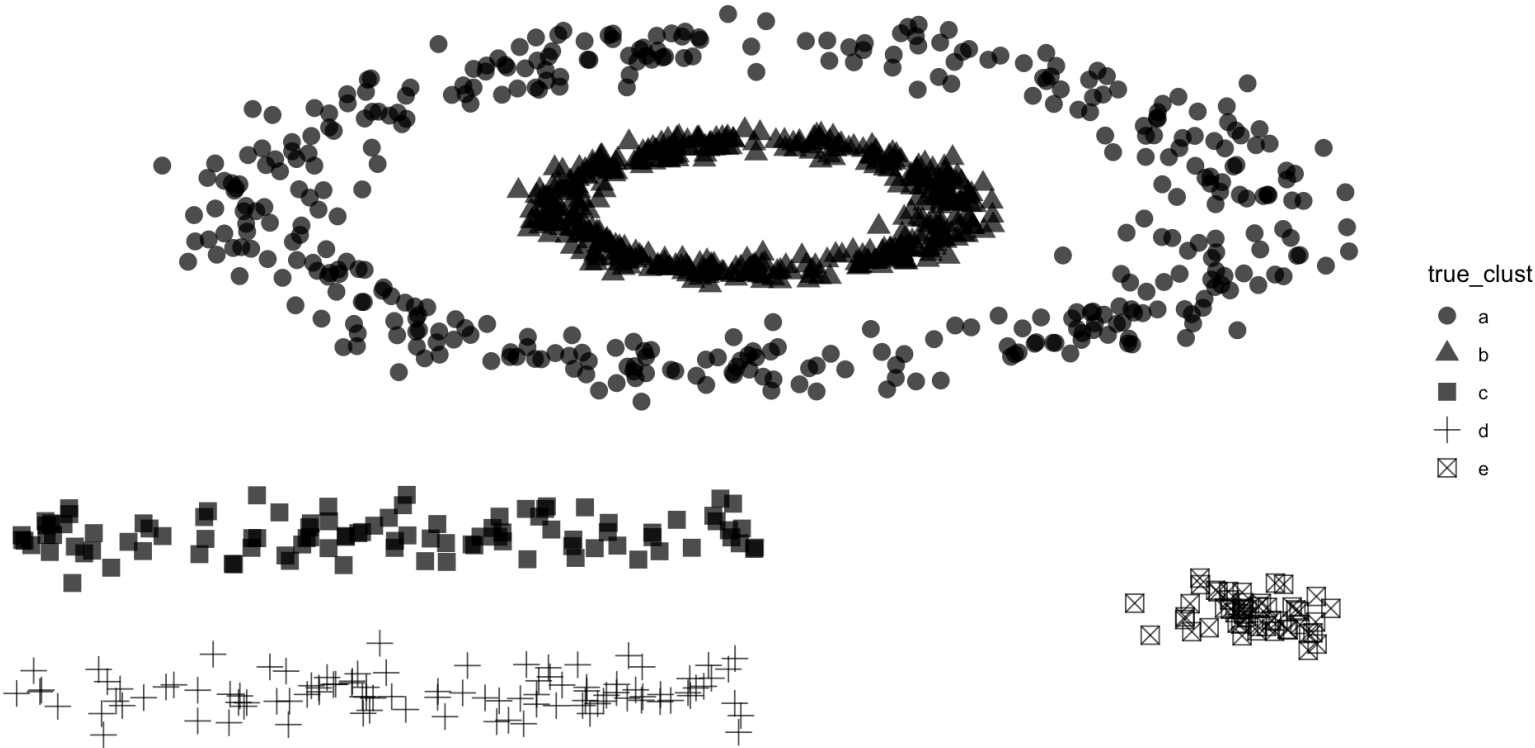


1. NJW; Ng, Jordan, & Weiss (2001)

clustering approaches

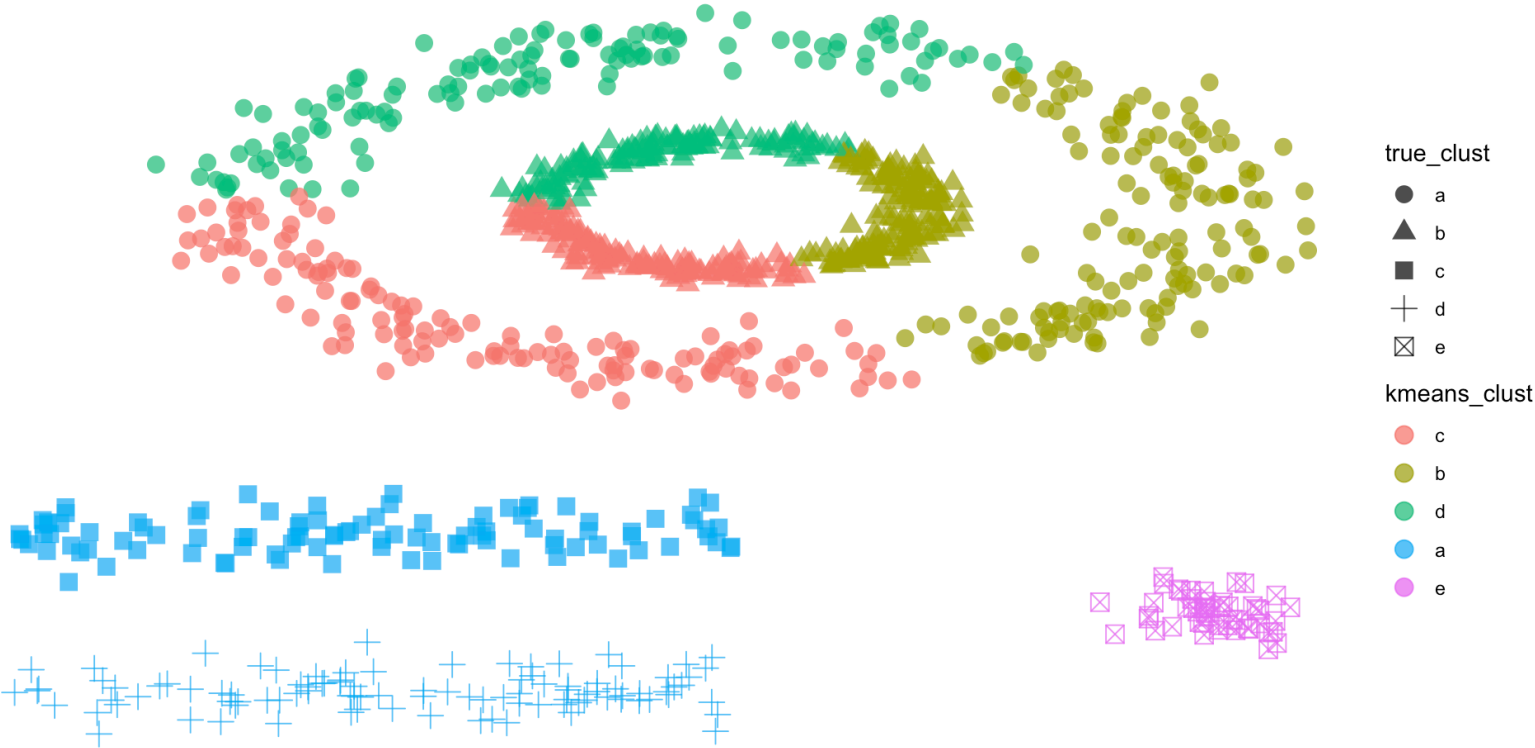
If a cluster structure does underlie the observations at hand, any approach will work

OR NOT?



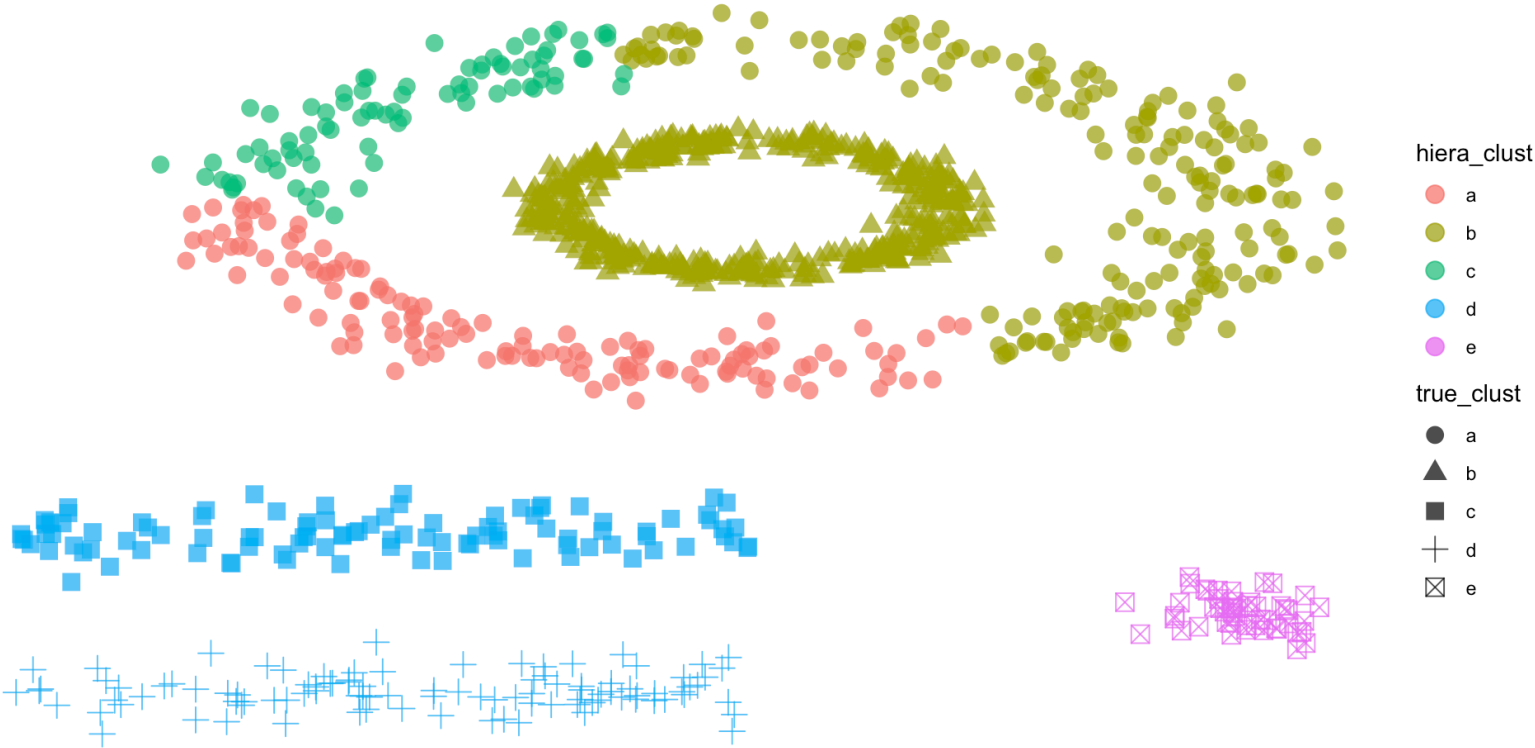
clustering approaches

partitioning
(Kmeans)



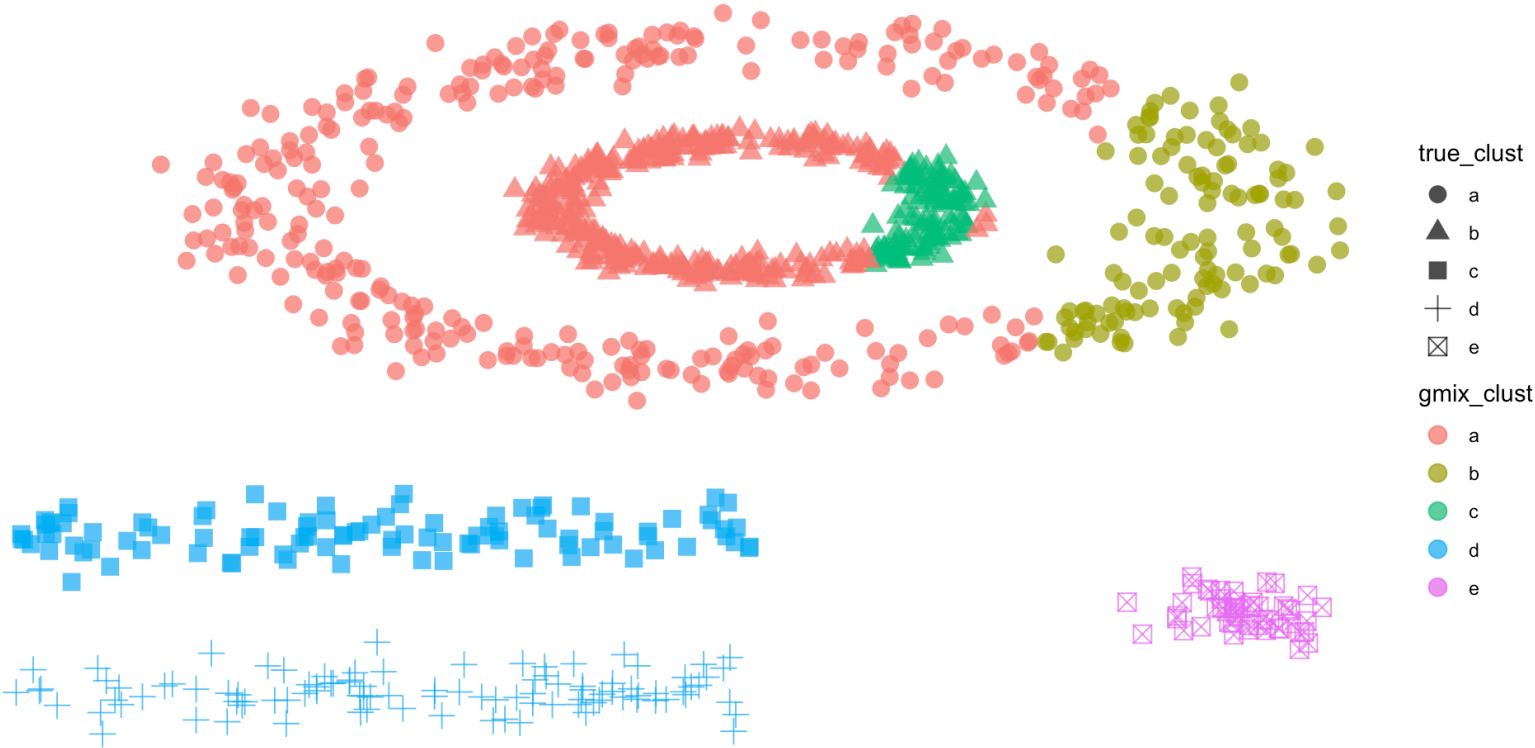
clustering approaches

agglomerative



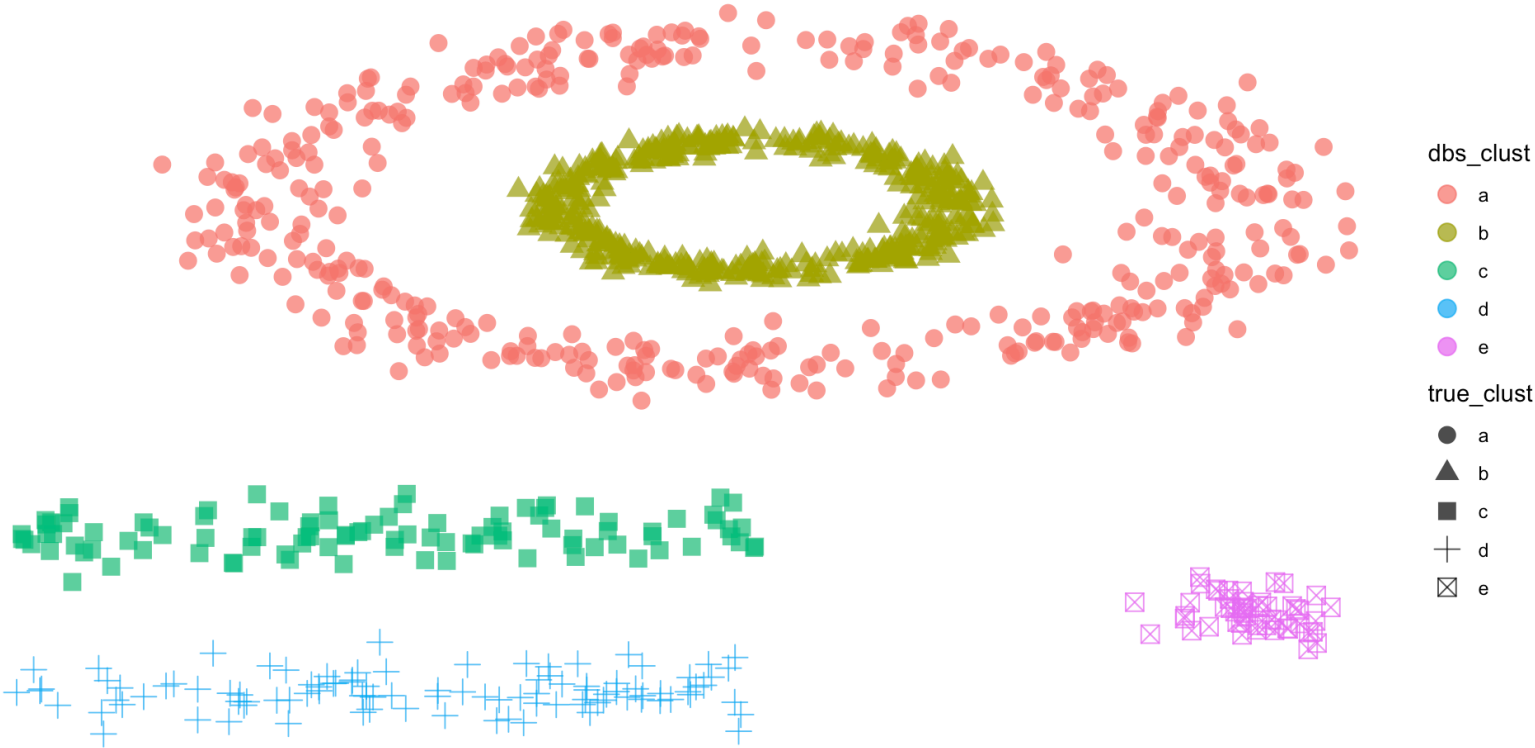
clustering approaches

model-based (GMM)



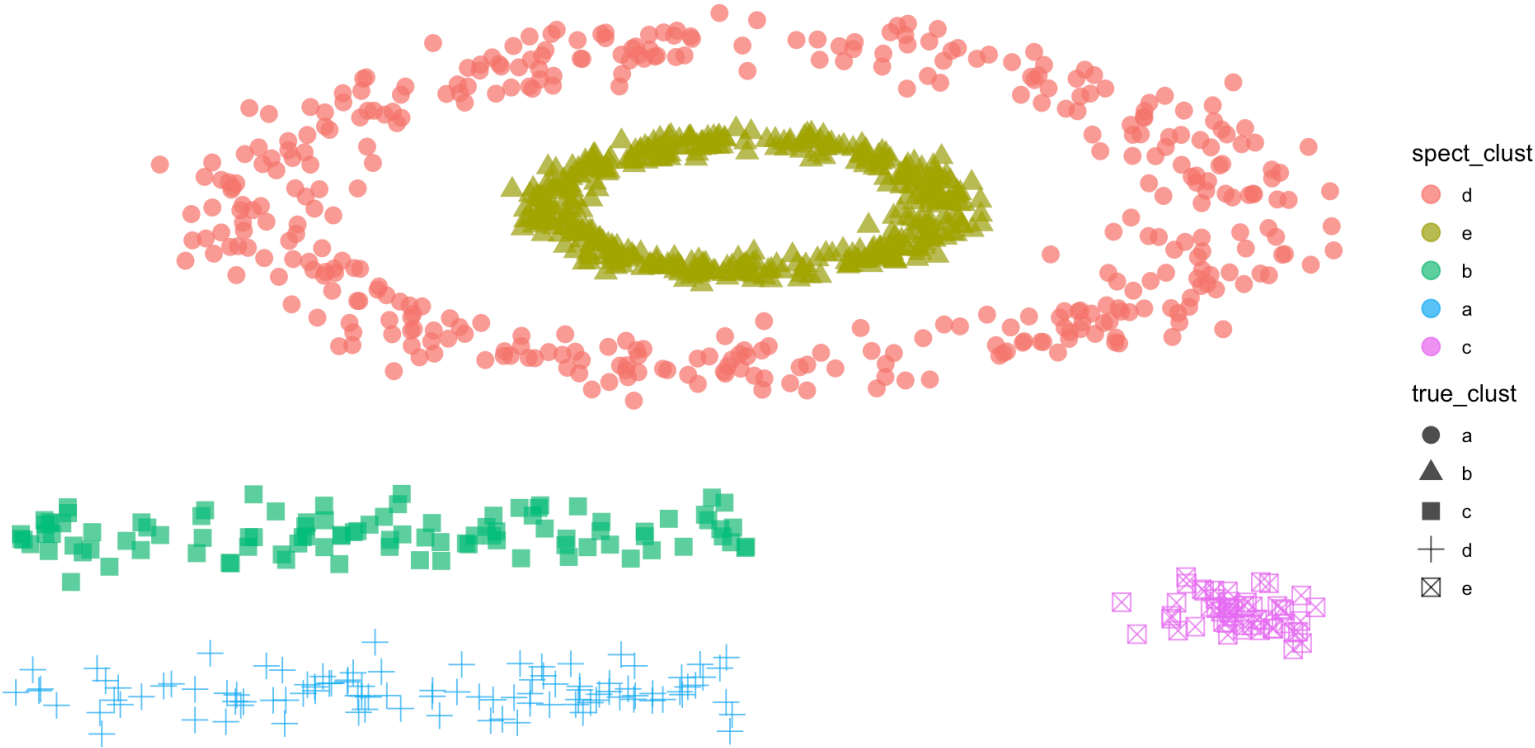
clustering approaches

density-based
(dbscan)



clustering approaches

spectral



spectral clustering in a nutshell

Spectral clustering: a graph partitioning problem

Graph representation

Represent the data matrix $\mathbf{X}$ as a graph and cut it into K groups, or clusters.

- observations are vertices;
- pairwise similarities are edge weights;
- large $w_{i\ell}$ indicates that observations i and ℓ should remain connected.

the distance determines the graph that is available to partition

From distances to affinities

Start from the $I \times I$ pairwise distance matrix $\mathbf{D}$ and construct

$$\mathbf{A} = \exp \left(-\frac{\mathbf{D}^2}{2\sigma^2} \right), \quad a_{ii} = 0.$$

Affinity

The elements $a_{i\ell}$ are high when observations i and ℓ are close under the chosen distance.

Neighbourhood scale

The parameter σ controls which pairwise relationships remain visible in the graph.

Making the graph easy to cut

An approximate solution to the graph partitioning problem¹:

Normalized graph Laplacian

$$\mathbf{L} = \mathbf{D}_r^{-1/2} \mathbf{A} \mathbf{D}_r^{-1/2} = \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^\top,$$

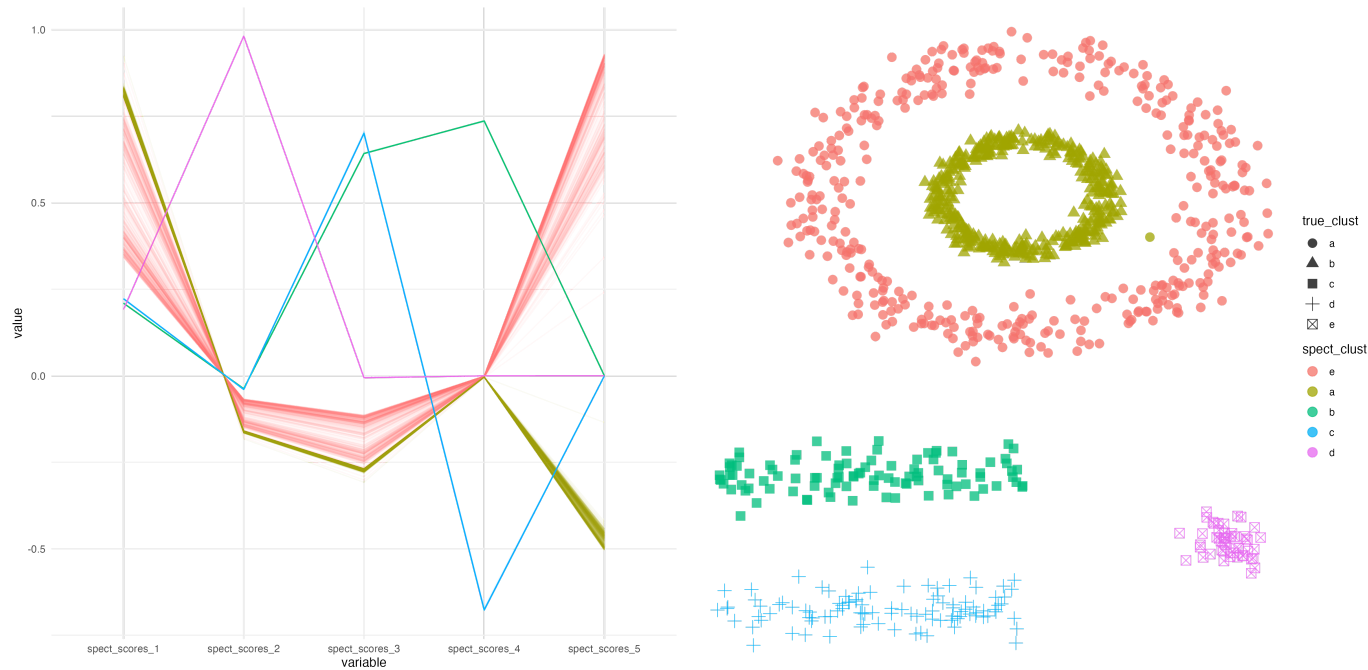
where $\mathbf{D}_r = \text{diag}(\mathbf{r})$, $\mathbf{r} = \mathbf{A}\mathbf{1}$, and $\mathbf{1}$ is an I -dimensional vector of ones.

Spectral embedding

Apply K -means to the rows of $\tilde{\mathbf{Q}}$, containing the first K eigenvectors of $\mathbf{L}$.

Why spectral clustering?

Spectral clustering works on the connectivity encoded by the graph and can recover non-convex structure.



feature-aware distances for mixed data

Mixed-data setup

A mixed data set

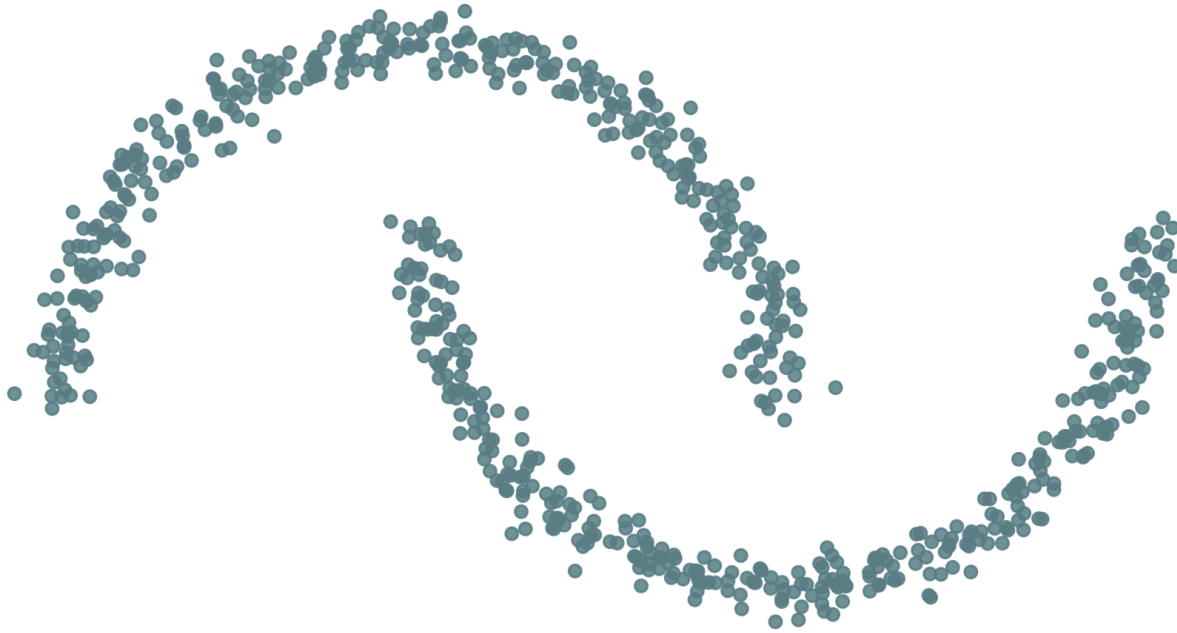
- I observations described by Q variables;
- Q_n numerical variables and Q_c categorical variables;
- the $I \times Q$ data matrix is column-wise partitioned:

$$\mathbf{X} = [\mathbf{X}_n, \mathbf{X}_c].$$

A general mixed distance between observations i and ℓ is

$$\begin{aligned} d(\mathbf{x}_i, \mathbf{x}_\ell) = & \sum_{j_n=1}^{Q_n} w_{j_n} \delta_{j_n}^n(\mathbf{x}_i^n, \mathbf{x}_\ell^n) \\ & + \sum_{j_c=1}^{Q_c} w_{j_c} \delta_{j_c}^c(\mathbf{x}_i^c, \mathbf{x}_\ell^c). \end{aligned}$$

The numerical geometry suggests two groups



2

connected shapes

A numerical distance can recover the **upper** and **lower** moon.

Geometry alone

There is no preferred subdivision within either moon.

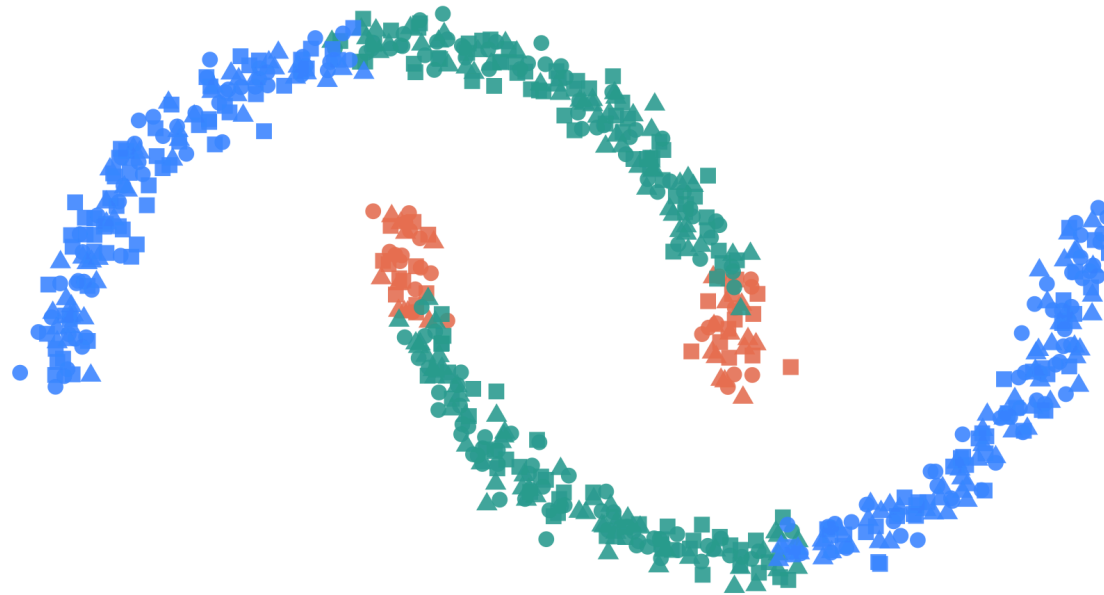
One factor is structured; the other is deliberately neutral

Signal band

- colour: A, B, C
- unequal arc lengths: 10%, 45%, 45%
- repeated on both moons

Nuisance factor

- shape: N_1, N_2, N_3
- balanced within every band
- spatially random



Signal band ● A ● B ● C Nuisance ● N1 ▲ N2 ■ N3

only the signal factor has a systematic relationship with the numerical geometry

The target is the moon **AND** the signal category

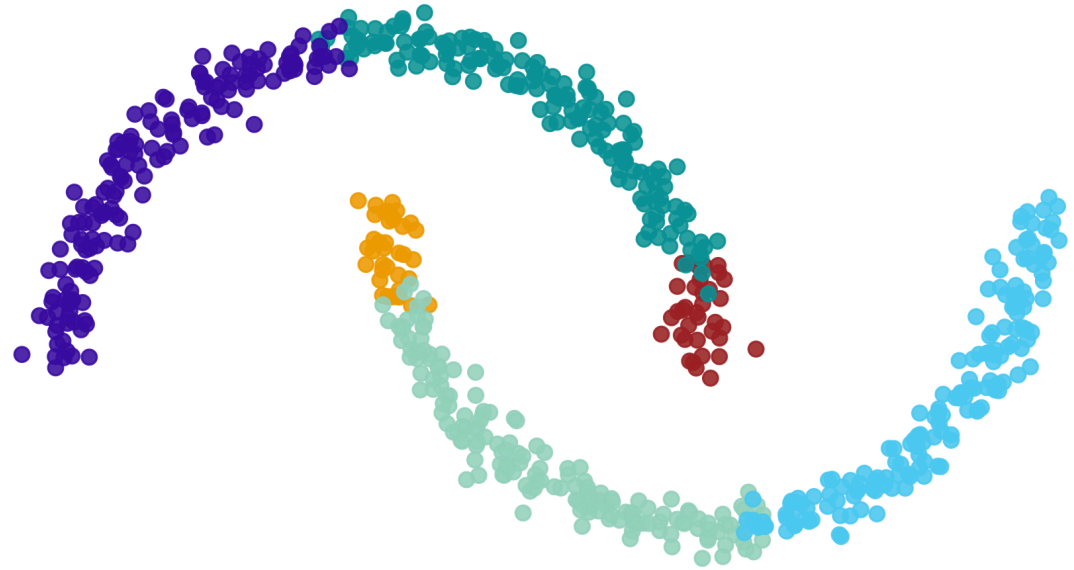
$$\text{cluster} = \text{moon} \times \text{band}$$

The desired partition contains six groups:

$UA, UB, UC,$
 $LA, LB, LC.$

The six-way intersection

Moon alone gives two groups. Band alone gives three.



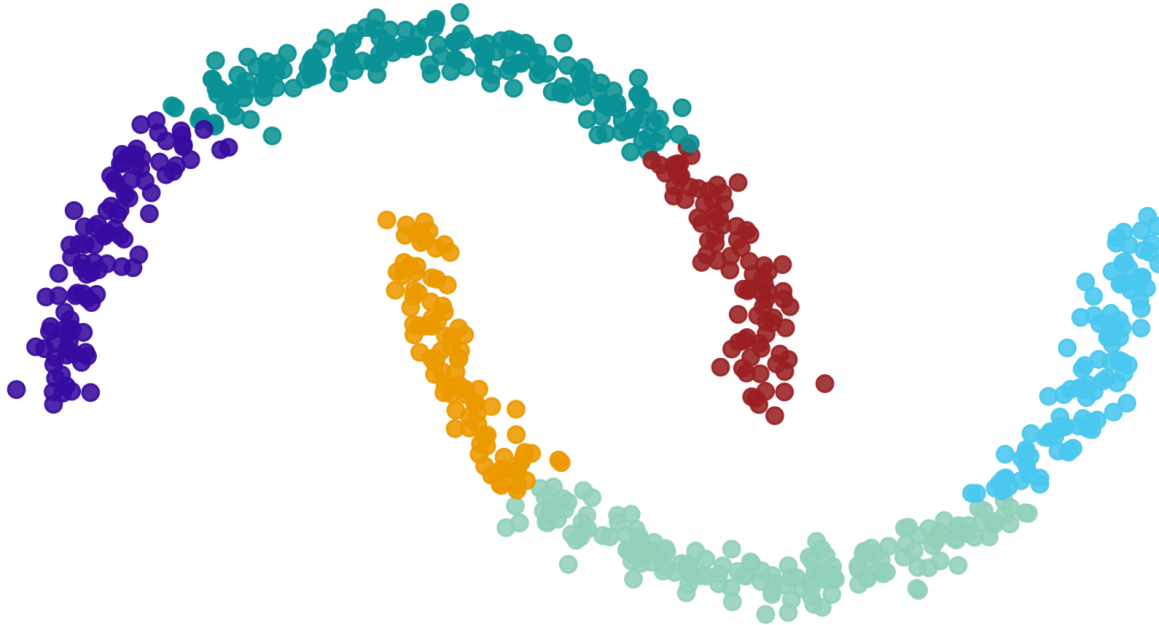
Without interaction, the graph follows geometry

ARI = 0.490

Spectral clustering is asked for $K = 6$, but the graph does not know where A , B , and C change.

The wrong boundaries

The moons are subdivided according to numerical geometry—not the signal factor.



With interaction, the graph recovers the six intersections

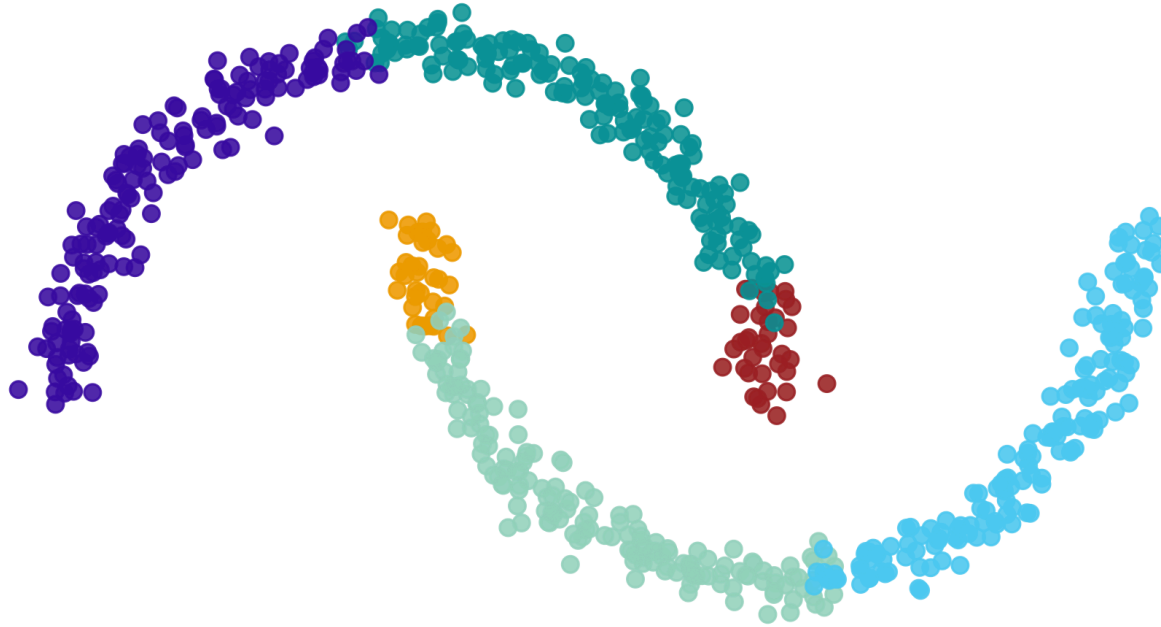
ARI = 1.000

The numerical distance preserves the two moons.

Δ_{int} adds barriers at the $A/B/C$ boundaries.

Exact recovery

The graph now encodes the required **moon AND category** structure.



The distance determines what spectral clustering can recover

numerical geometry + categorical association
+ cross-type interaction $\implies$ graph structure

If clusters are defined jointly by numerical and categorical information,
the distance must make that joint structure visible **before** the graph is partitioned.

this motivates an interaction-aware construction for mixed-data dissimilarities

Independence-based distances

Independence-based pairwise distance

No inter-variable relations are considered.

- continuous variables: **Euclidean** or **Manhattan** distances;
- categorical variables: **Hamming** / matching distance, among many others;
- mixed data: **Gower** dissimilarity index.

Beyond commensurability

Commensurability makes variable contributions comparable across **scales** and data **types**¹.

If variables are correlated or associated, shared information may still enter the distance repeatedly: **redundancy**.

The graph inherits the feature representation

Scale and type

Are contributions comparable across measurement scales and data types?

Association

Is shared information within the numerical or categorical block counted repeatedly?

Interaction

Do numerical features make some categorical distinctions more meaningful than others?

feature-based interpretation starts with the components used to construct $\mathbf{D}$

association-based distances

Continuous features: Mahalanobis distance

Association-aware continuous component

Let $\mathbf{X}_n$ be an $I \times Q_n$ matrix and let $\mathbf{S}$ be its sample covariance matrix. The Mahalanobis distance matrix is

$$\mathbf{D}_{mah} = [\text{diag}(\mathbf{G})\mathbf{1}^\top + \mathbf{1} \text{diag}(\mathbf{G})^\top - 2\mathbf{G}]^{\odot 1/2},$$

where

$$\mathbf{G} = (\mathbf{C}\mathbf{X}_n)\mathbf{S}^{-1}(\mathbf{C}\mathbf{X}_n)^\top, \quad \mathbf{C} = \mathbf{I} - \frac{1}{I}\mathbf{1}\mathbf{1}^\top.$$

correlations are represented through $\mathbf{S}^{-1}$ rather than counted as separate independent directions

Categorical features: the delta framework

Let $\mathbf{X}_c$ be an $I \times Q_c$ categorical data matrix, and let

$$\mathbf{Z} = [\mathbf{Z}_1, \dots, \mathbf{Z}_{Q_c}]$$

be its one-hot encoding.

$$\mathbf{D}_c = \mathbf{Z}\Delta\mathbf{Z}^\top, \quad \Delta = \text{bdiag}(\Delta_1, \dots, \Delta_{Q_c}).$$

Setting the category-dissimilarity blocks Δ_j determines the categorical distance measure¹.

category-level distinctions are translated into observation-level distances

From co-occurrences to conditional profiles

The empirical co-occurrence matrix is

$$\mathbf{P} = \frac{1}{I} \mathbf{Z}^\top \mathbf{Z}.$$

Its off-diagonal blocks contain joint proportions between categorical variables.

Let

$$\mathbf{P}_z = \mathbf{P} \odot \mathbf{I}_{Q^*}.$$

The block matrix

$$\mathbf{R} = \mathbf{P}_z^{-1} (\mathbf{P} - \mathbf{P}_z)$$

contains conditional profiles across variables.

Total variation distance between categories

Let $\mathbf{r}_a^{ji}$ and $\mathbf{r}_b^{ji}$ be rows of the (j, i) th off-diagonal block of $\mathbf{R}$.

Association-aware categorical component

For categories a and b of variable j ,

$$\begin{aligned}\delta_{tvd}^j(a, b) &= \sum_{i \neq j}^{Q_c} w_{ji} \Phi^{ji} \left(\mathbf{r}_a^{ji}, \mathbf{r}_b^{ji} \right) \\ &= \sum_{i \neq j}^{Q_c} w_{ji} \left[\frac{1}{2} \sum_{\ell=1}^{q_i} \left| \mathbf{r}_{a\ell}^{ji} - \mathbf{r}_{b\ell}^{ji} \right| \right],\end{aligned}$$

with $w_{ji} = 1/(Q_c - 1)$ under equal weighting¹.

From category dissimilarities to $\mathbf{D}_{tvd}$

Collect the category-dissimilarity matrices in

$$\Delta^{(tvd)} = \text{bdiag} \left(\Delta_1^{(tvd)}, \dots, \Delta_{Q_c}^{(tvd)} \right).$$

Then

$$\begin{aligned} \mathbf{D}_{tvd} &= \mathbf{Z} \Delta^{(tvd)} \mathbf{Z}^\top \\ &= \sum_{j=1}^{Q_c} \mathbf{z}_j \Delta_j^{(tvd)} \mathbf{z}_j^\top. \end{aligned}$$

categorical mismatches are weighted by how differently their categories relate to the other categorical features

Association-aware mixed representation

Within each variable block:

Numerical block

$\mathbf{D}_{mah}$ accounts for covariance and redundancy among continuous variables.

Categorical block

$\mathbf{D}_{tvd}$ accounts for associations among categorical variables.

$$\mathbf{D}_{mix}^{(assoc)} = \mathbf{D}_{mah} + \mathbf{D}_{tvd}.$$

within-block structure is represented, but cross-type structure is still missing

interaction-aware extensions

Cross-type interactions

Association-aware distances account for relations

- among continuous variables;
- among categorical variables.

Cross-type structure

In mixed data, categorical differences may be meaningful because they are reflected in the continuous variables.

the next step is to make the distance **interaction-aware**

How to measure interactions¹

Define Δ^{int} to encode continuous–categorical interactions and augment Δ^{tvd} :

$$\tilde{\Delta} = (1 - \alpha)\Delta^{tvd} + \alpha\Delta^{int}, \quad \alpha = \frac{1}{Q_c}.$$

The mixed dissimilarity becomes

$$\mathbf{D}_{mix}^{(int)} = \mathbf{D}_{mah} + \mathbf{D}_{cat}^{(int)}, \quad \mathbf{D}_{cat}^{(int)} = \mathbf{Z}\tilde{\Delta}\mathbf{Z}^\top.$$

What is Δ^{int} ?

The entry $\delta_{int}^j(a, b)$ measures how much the continuous variables help discriminate between observations choosing category a and those choosing category b for categorical variable j .

Category-pair classification problem

For each pair (a, b) :

- use the continuous variables as predictors;
- classify observations belonging to categories a and b ;
- use a nearest-neighbour rule in the continuous space.

Computing Δ_j^{int}

For each categorical variable j and each category pair (a, b) :

1. use $\mathbf{D}_{mah}$ to identify neighbours among observations belonging to a or b ;
2. consider a proportion of neighbours, say $\pi_{nn} = 0.1$;
3. classify observations using a prior-corrected decision rule;
4. compute balanced accuracy.

$$\text{BAcc}^j(a, b) = \frac{1}{2} \left[\text{TPR}_a^j(a, b) + \text{TPR}_b^j(a, b) \right].$$

Map predictive performance onto a non-negative separability scale:

$$\delta_{int}^j(a, b) = \max \{0, 2 \text{BAcc}^j(a, b) - 1\} \in [0, 1].$$

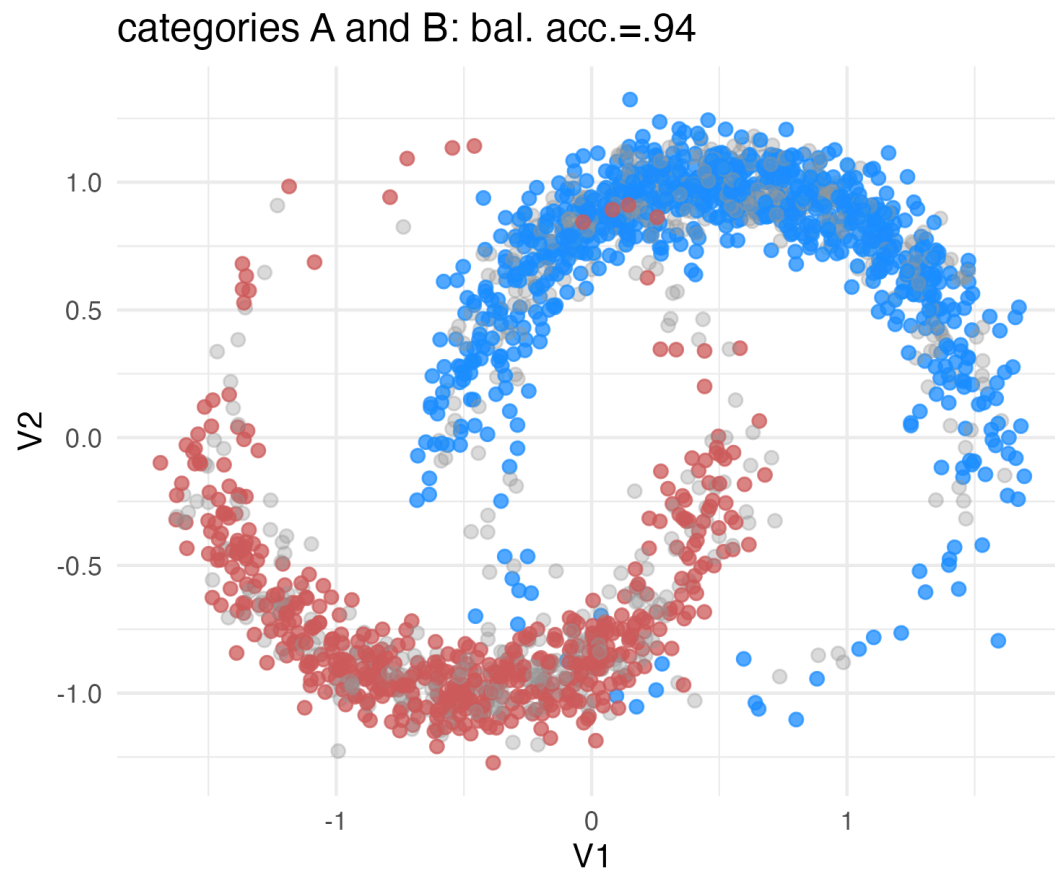
0 means chance-level or worse; 1 means perfect separation

Computing Δ_j^{int}

For categorical variable j with q_j categories, compute $\frac{q_j(q_j-1)}{2}$ category-pair quantities.

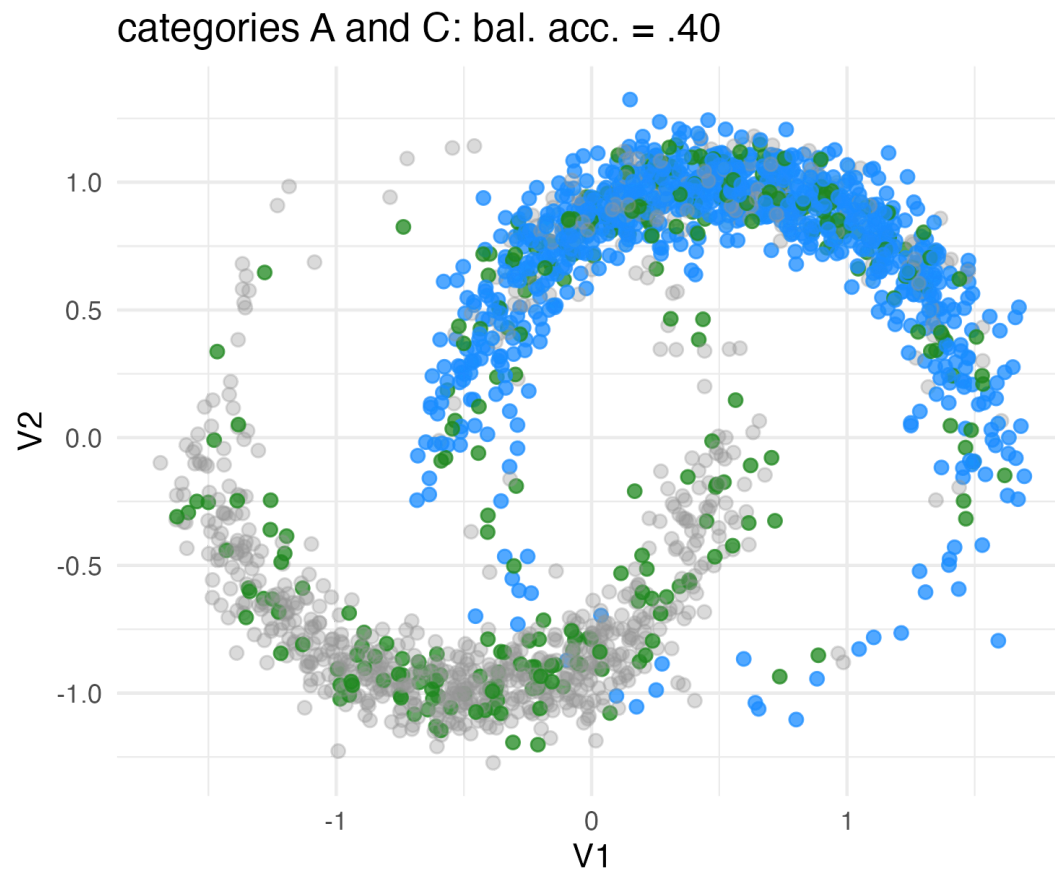
$$\Delta_{int} = \begin{pmatrix} 0 & \cdot & \cdot & \cdot \\ \cdot & 0 & \cdot & \cdot \\ \cdot & \cdot & 0 & \cdot \\ \cdot & \cdot & \cdot & 0 \end{pmatrix}$$

Computing Δ_j^{int}



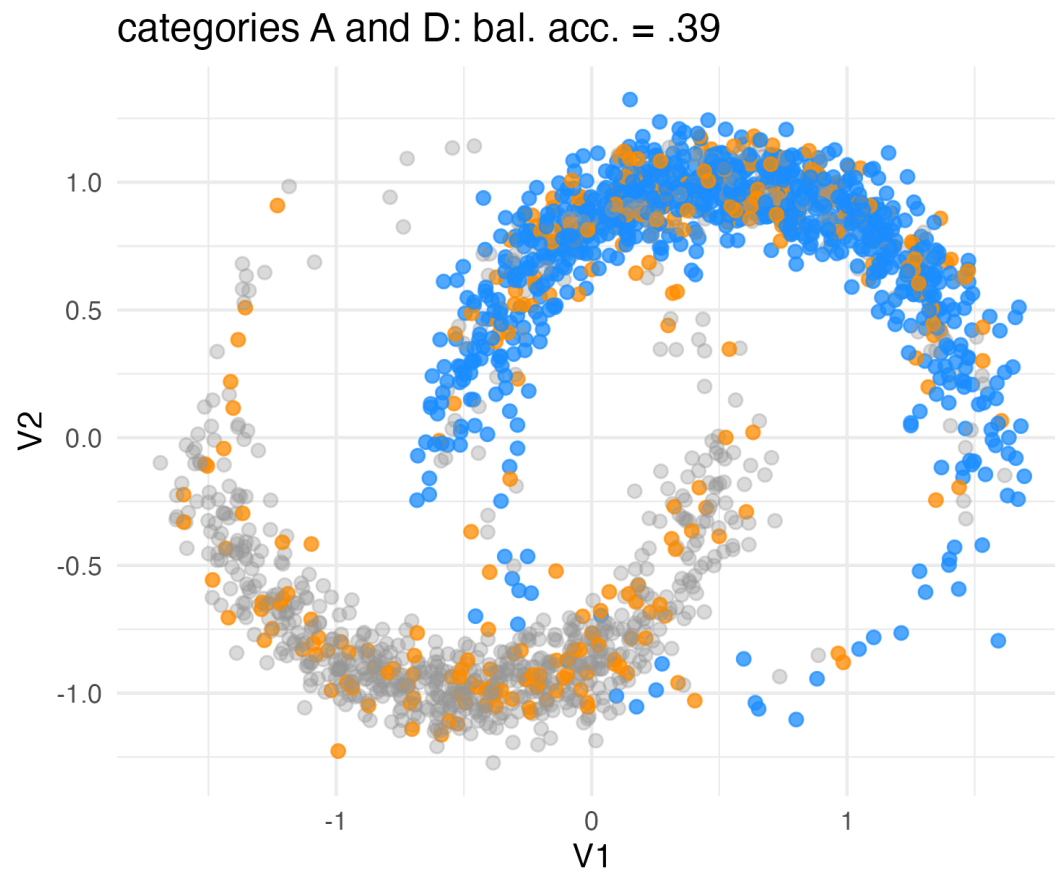
$$\Delta_{int} = \begin{pmatrix} 0 & 0.94 & \cdot & \cdot \\ 0.94 & 0 & \cdot & \cdot \\ \cdot & \cdot & 0 & \cdot \\ \cdot & \cdot & \cdot & 0 \end{pmatrix}$$

Computing Δ_j^{int}



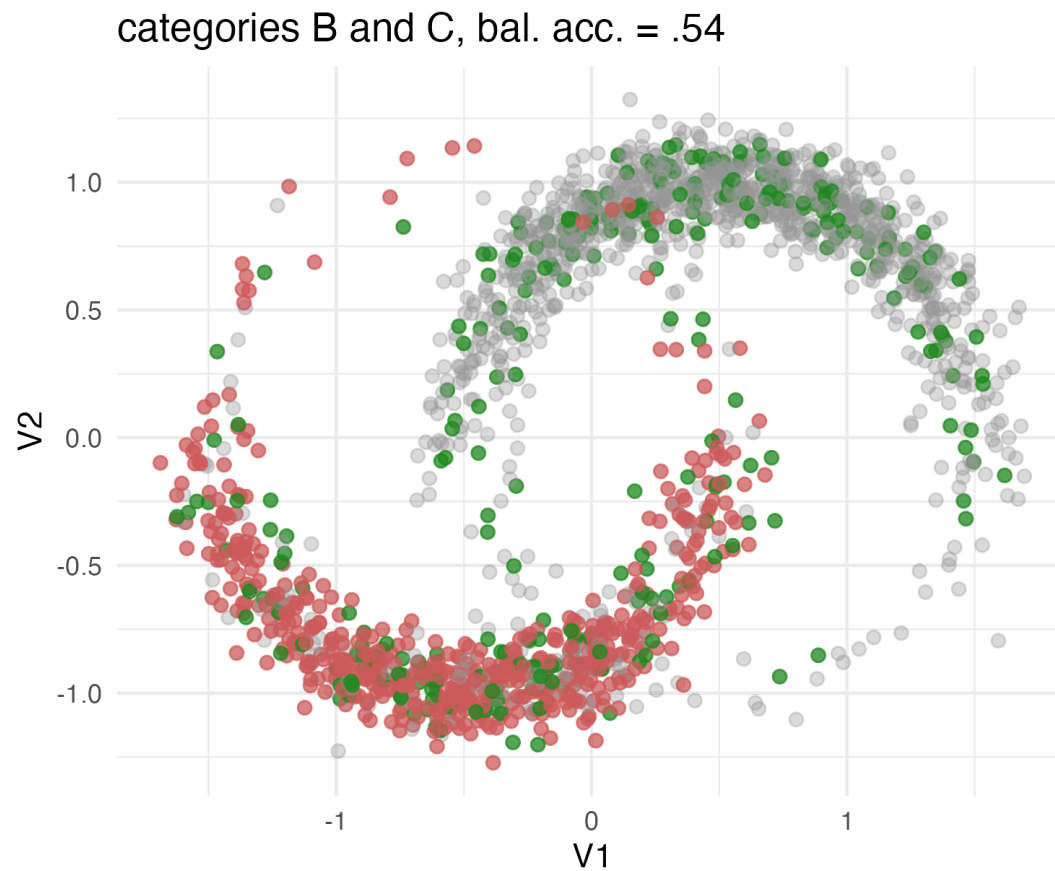
$$\Delta_{int} = \begin{pmatrix} 0 & 0.94 & \textcolor{red}{0.40} & \cdot \\ 0.94 & 0 & \cdot & \cdot \\ \textcolor{red}{0.40} & \cdot & 0 & \cdot \\ \cdot & \cdot & \cdot & 0 \end{pmatrix}$$

Computing Δ_j^{int}



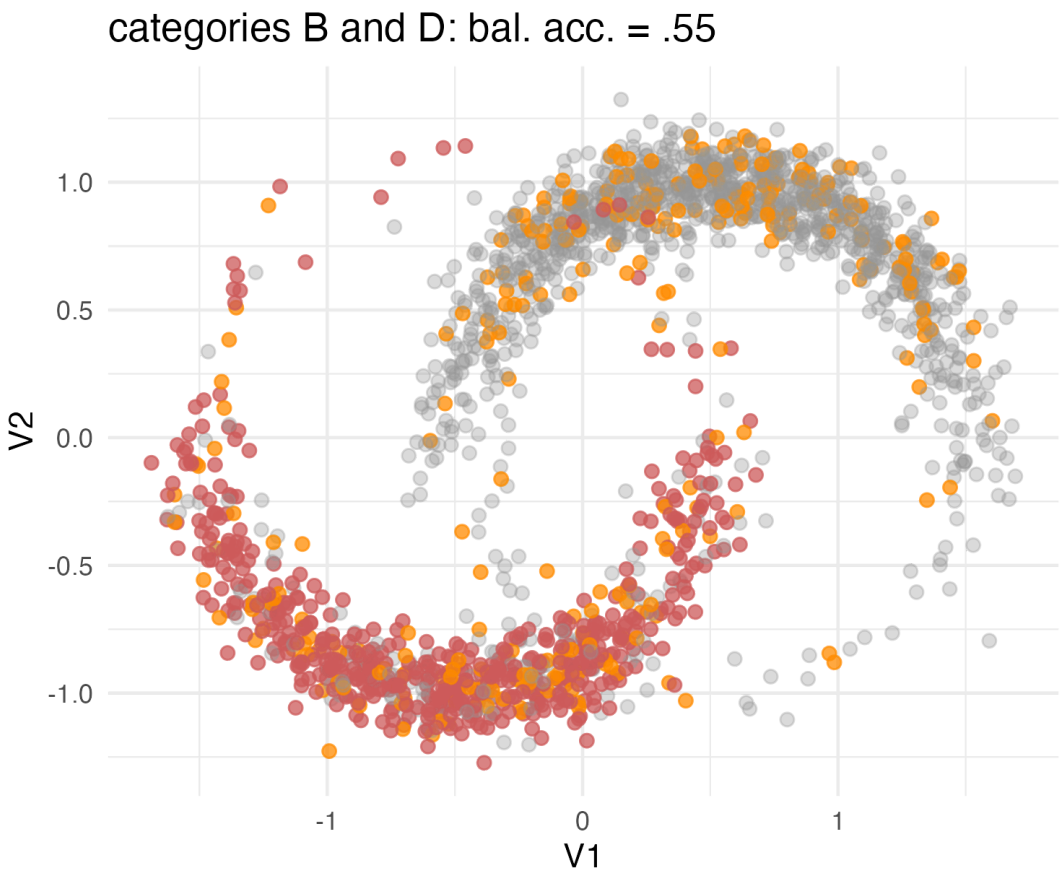
$$\Delta_{int} = \begin{pmatrix} 0 & 0.94 & 0.40 & 0.39 \\ 0.94 & 0 & \cdot & \cdot \\ 0.40 & \cdot & 0 & \cdot \\ 0.39 & \cdot & \cdot & 0 \end{pmatrix}$$

Computing Δ_j^{int}



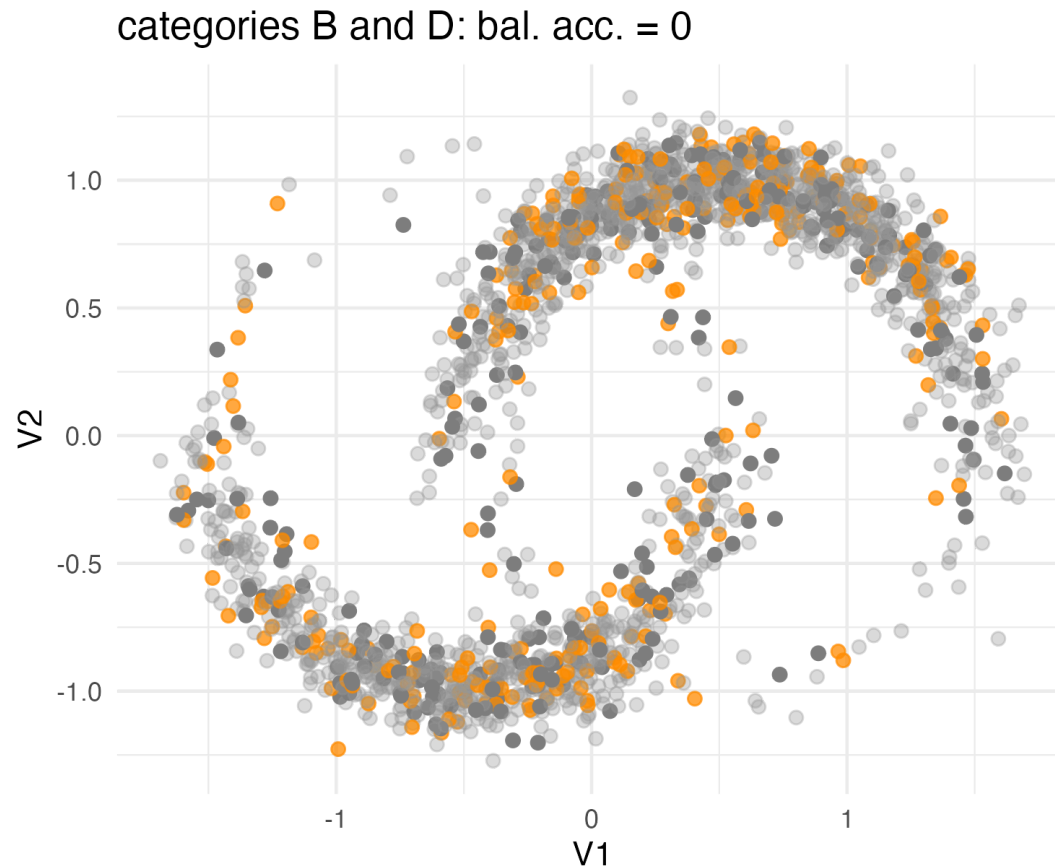
$$\Delta_{int} = \begin{pmatrix} 0 & 0.94 & 0.40 & 0.39 \\ 0.94 & 0 & 0.54 & \cdot \\ 0.40 & 0.54 & 0 & \cdot \\ 0.39 & \cdot & \cdot & 0 \end{pmatrix}$$

Computing Δ_j^{int}



$$\Delta_{int} = \begin{pmatrix} 0 & 0.94 & 0.40 & 0.39 \\ 0.94 & 0 & 0.54 & 0.55 \\ 0.40 & 0.54 & 0 & . \\ 0.39 & 0.55 & . & 0 \end{pmatrix}$$

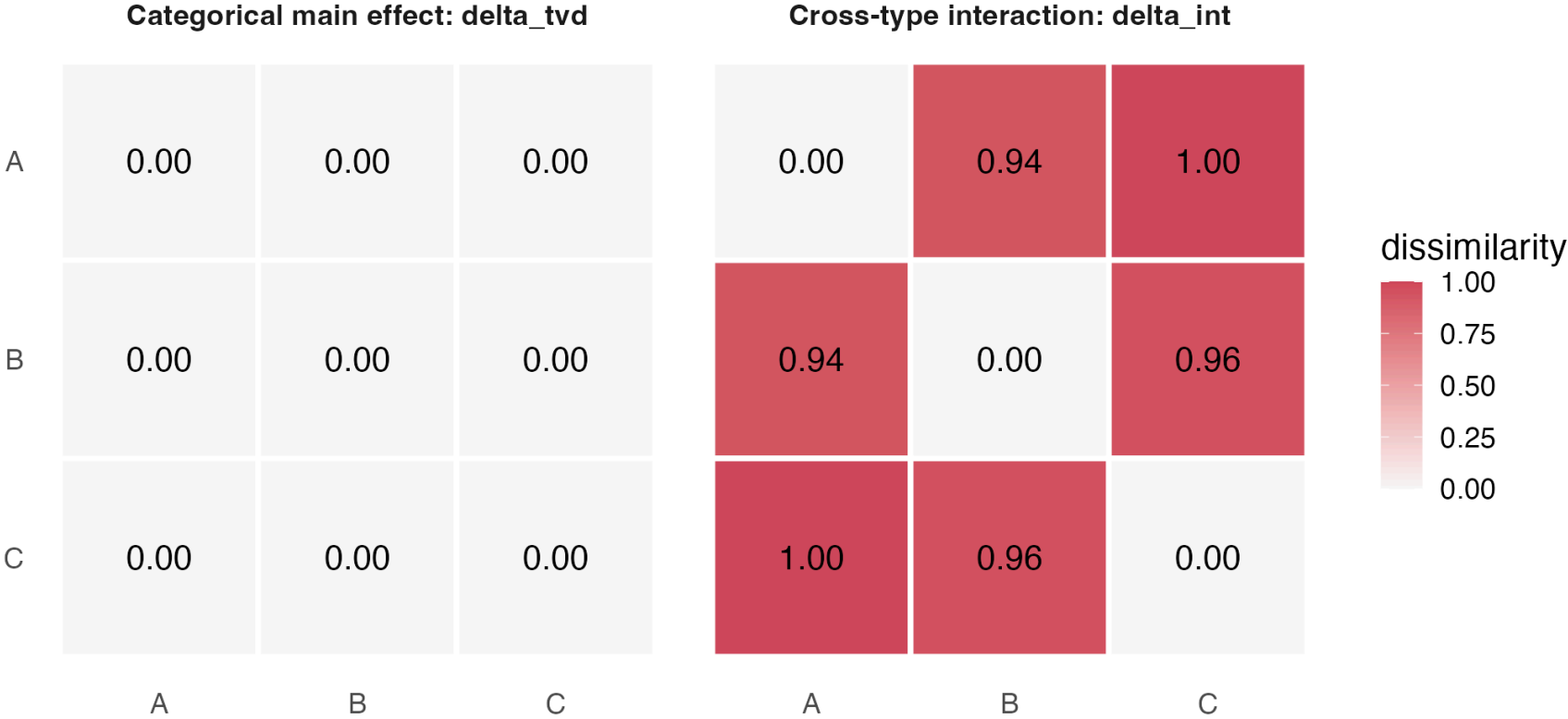
Computing Δ_j^{int}



$$\Delta_{int} = \begin{pmatrix} 0 & 0.94 & 0.40 & 0.39 \\ 0.94 & 0 & 0.54 & 0.55 \\ 0.40 & 0.54 & 0 & 0 \\ 0.39 & 0.55 & 0 & 0 \end{pmatrix}$$

summarize category-pair separability in the continuous space

Back to the motivating example



$$\Delta_{\text{tvd}} = 0$$

Band and nuisance are independent.

$$\delta^{\text{int}}(a, b) = \max\{0, 2 \text{BAcc}(a, b) - 1\}$$

Numerical neighbours discriminate every band pair.

$$\delta^{\text{int}}_{AB} = 0.94 \quad \delta^{\text{int}}_{AC} = 1.00 \quad \delta^{\text{int}}_{BC} = 0.96$$

A feature-based decomposition of pairwise dissimilarity

For observations i and ℓ ,

$$d_{mix}^{(int)}(i, \ell) = d_{mah}(\mathbf{x}_{i_n}, \mathbf{x}_{\ell_n}) + \sum_{j=1}^{Q_c} \left[(1 - \alpha) \delta_{tvd}^j(x_{ij_c}, x_{\ell j_c}) + \alpha \delta_{int}^j(x_{ij_c}, x_{\ell j_c}) \right].$$

every graph edge is built from continuous association, categorical association, and cross-type interaction components

The additive construction is coherent

Define the maximum-normalised continuous and categorical kernels:

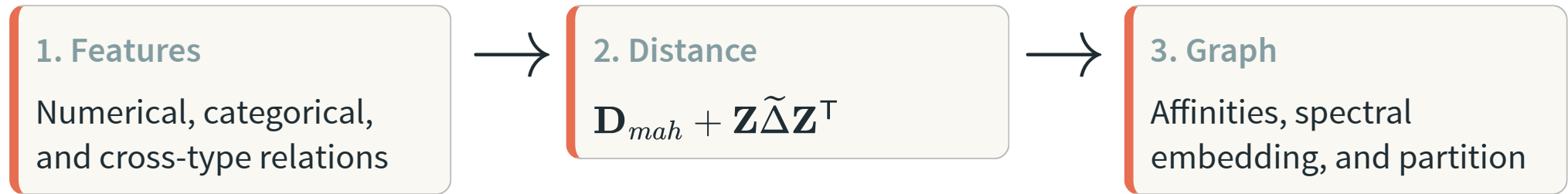
$$\overline{K}_n(i, \ell) = \exp \{ -d_{mah}(\mathbf{x}_{i_n}, \mathbf{x}_{\ell_n}) \},$$

$$\overline{K}_c^{(int)}(i, \ell) = \exp \left\{ -d_c^{(int)}(\mathbf{x}_{i_c}, \mathbf{x}_{\ell_c}) \right\}.$$

Their product satisfies

$$-\log \left[\overline{K}_n(i, \ell) \overline{K}_c^{(int)}(i, \ell) \right] = d_{mix}^{(int)}(i, \ell).$$

From feature relations to a spectral solution



interpreting the solution means tracing which feature relations made observations close

synthetic experiments

Simulation 1: associated mixed variables

General design

- $I = 1000$ and $I = 2000$;
- $K = 3$ clusters;
- 50 datasets per configuration;
- spectral clustering applied to each dissimilarity matrix;
- performance measured by adjusted Rand index.

Feature structure

- correlated numerical variables;
- associated categorical variables;
- categorical variables depending on numerical variables;
- an additional noise scenario.

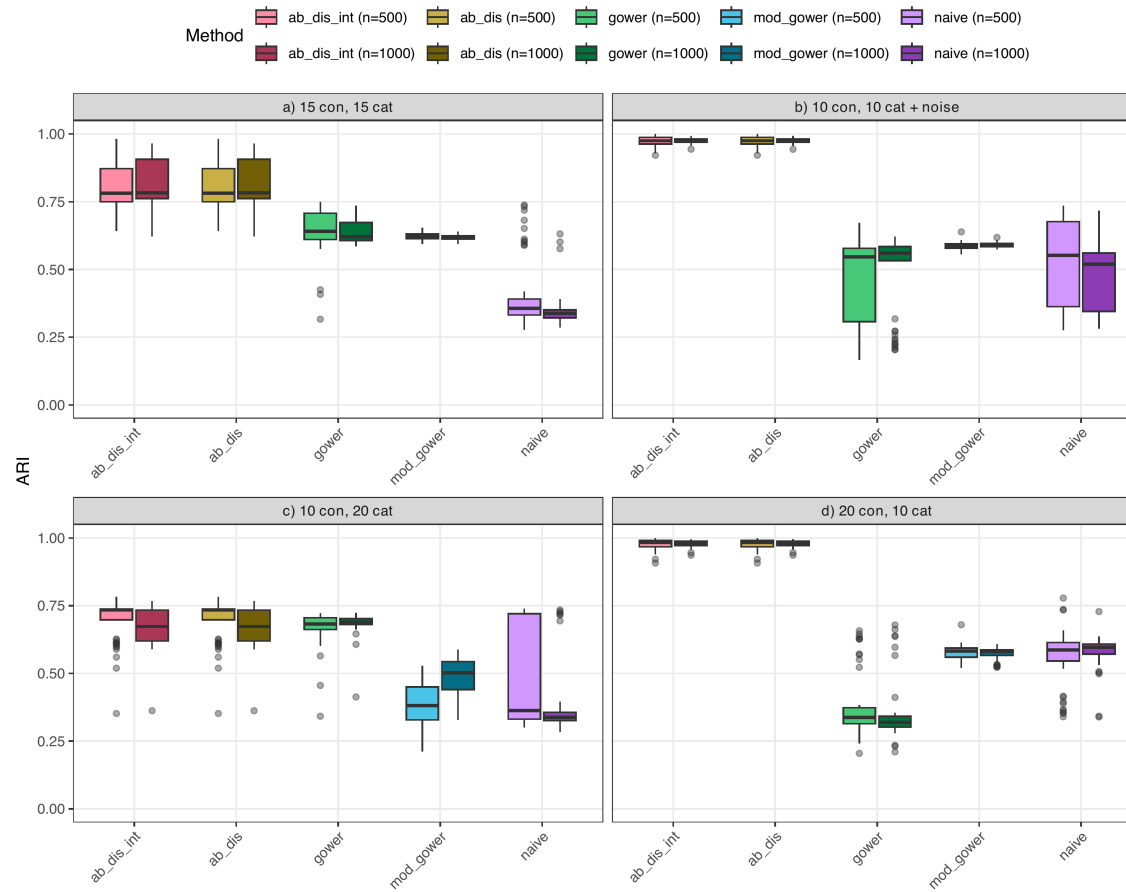
Simulation 1: variable composition

Four mixed-data scenarios:

1. $Q_n = 15, Q_c = 15$;
2. $Q_n = 10, Q_c = 20$;
3. $Q_n = 20, Q_c = 10$;
4. $Q_n = 10, Q_c = 10$, plus noise variables.

the design induces numerical correlations, categorical associations, and numerical–categorical dependence

Simulation 1: results



- Association-based distances are generally the most accurate.
- When continuous variables dominate, association-based distances achieve almost perfect recovery.
- Standard mixed-data distances are more sensitive to variable composition and noise.

Simulation 2: interaction-driven clusters

Design

- $I = 500$ and $I = 1000$;
- six continuous and three categorical variables;
- V_4, V_5, V_6 generated independently from $N(0, 1)$;
- V_1, V_2, V_3 generated conditionally on C_1 and C_2 .

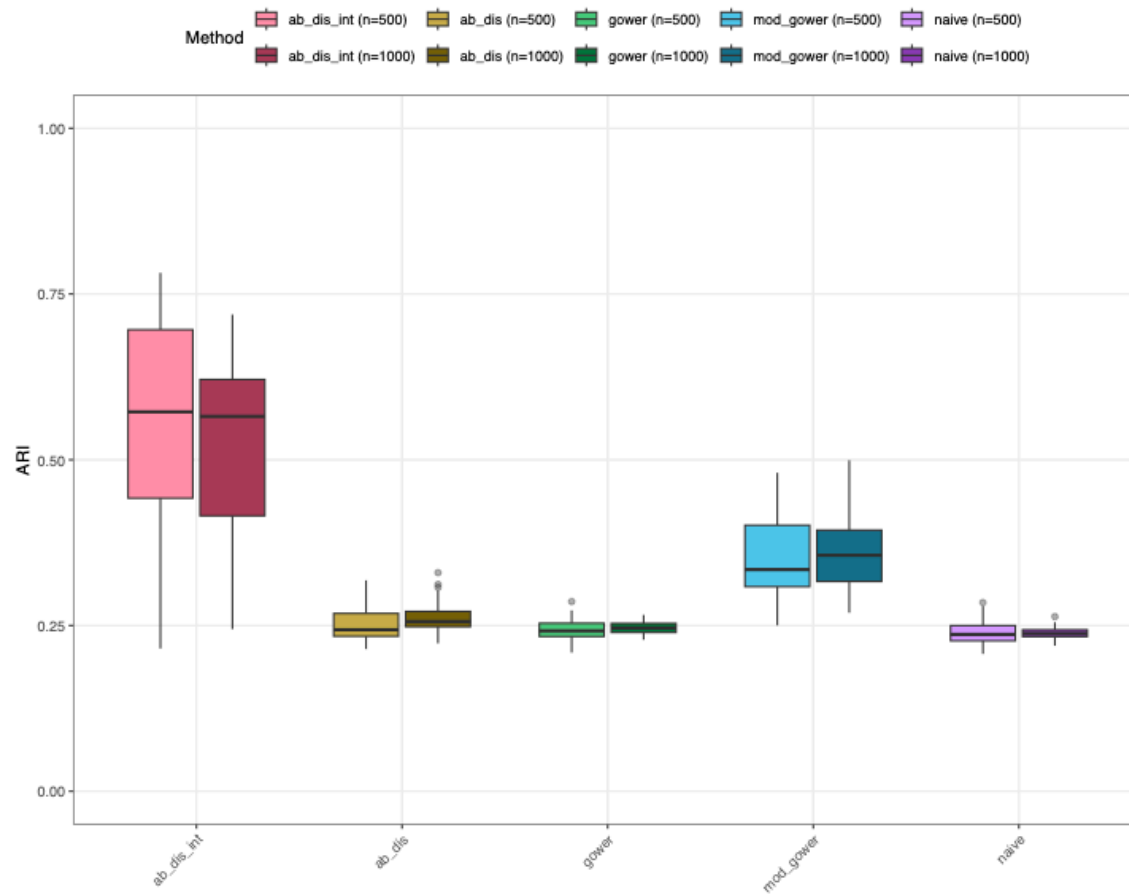
Main feature

The clusters are not defined by the continuous variables alone or the categorical variables alone.

They are defined by their **cross-type interaction**.

this is the setting where Δ^{int} should matter

Simulation 2: results



- `ab_dis_int` outperforms the competitors.
- Distances that do not account for cross-type structure provide almost chance-level performance.

interpretation and conclusions

Reading a spectral solution through its features

D_{mah}

Which numerical differences remain after accounting for covariance?

Δ^{tvd}

Which categorical distinctions are supported by their relations with other categorical features?

Δ^{int}

Which categorical distinctions are expressed in the continuous feature space?

these components explain the representation supplied to spectral clustering—not a post-hoc relabelling of the clusters

What the experiments support

Association structure matters

Mahalanobis and TVD-based components account for redundancy within the numerical and categorical blocks.

Cross-type interactions must be encoded

Δ^{int} translates category separability in the numerical space into categorical dissimilarities.

The distance determines what spectral clustering can recover

The simulations favour association-aware distances and show a clear gain from Δ^{int} when clusters are interaction-driven.

Main message

A spectral clustering solution is only as informative as the pairwise representation supplied to it.

For mixed data, feature-based interpretation requires distances that make scale, association, and cross-type interaction explicit.

Main references

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